

Tool for Cable Fault Detection - EDF

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Abstract

Electrical cable faults caused by corrosion, mechanical stress, or external damage pose significant economic and safety risks in power distribution networks. Existing detection methods often require expensive equipment and expert signal interpretation, limiting their accessibility.

This project develops an interactive simulation tool for cable fault detection based on reflectometry principles. A cable is modeled as a distributed-parameter transmission line; localized impedance discontinuities are introduced to represent faults, and the resulting reflected signals are simulated in both time and frequency domains. FDTD schemes are implemented and compared, alongside a frequency-domain analysis capturing standing wave patterns and impedance profiles. The tool is deployed as an interactive web application, allowing users to configure cable parameters and fault properties without signal processing expertise. Fault localization accuracy, simulation stability, and ease of use are evaluated across several test scenarios.

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Introduction

Electrical cables are the backbone of power distribution networks, and their reliable operation is critical to both economic continuity and public safety. In practice, cables are subject to degradation from multiple sources: corrosion progressively weakens conductor integrity, mechanical stress from ground movement or installation constraints introduces structural damage, and external factors such as animal interference can cause localized insulation failures. When a fault occurs in an underground cable network, locating it precisely is a significant challenge; the fault may be anywhere along several kilometers of buried infrastructure, invisible to direct inspection.

0.1 State of the Art

Current detection approaches rely on reflectometry-based measurement devices, which inject a signal into the cable and analyze the reflected response to infer the fault position. While effective, these instruments require trained operators to interpret the resulting waveforms, and are not always well suited to educational or exploratory use; the main limitations being measurement accuracy, which depends on the technique and setup, and the difficulty of interpreting the resulting signals. Engineers and technicians currently lack accessible tools to simulate cable faults, visualize signal behavior, or test detection hypotheses without deploying physical equipment. Beyond detection itself, EDF is actively working on training AI algorithms to predict the source of cable faults, a task that requires large labeled datasets, which a simulation tool can help generate.

The academic literature has established a solid theoretical foundation for reflectometry-based fault detection, covering both time-domain and frequency-domain approaches. Research in this area has produced methods capable of highly precise fault localization on simple cables and complex branched networks alike. The key practical challenges identified by this body of work include interference between the diagnostic signal and the cable's normal operating signals and degradation of localization accuracy caused by signal attenuation and dispersion over long cable runs.

On the industrial side, analogous efforts in the power sector illustrate both the potential and the current limitations of

simulation-based cable diagnostics. Interactive simulation applications have been developed that allow field technicians to enter cable data, select a fault type, and modify an underlying model on the fly to assess whether a cable needs to be repaired or replaced [3]. This demonstrates that simulation tools can meaningfully improve the efficiency of field maintenance work even in demanding industrial environments. However, such tools are typically built around proprietary solvers, require significant domain expertise to configure, and are not designed to expose the underlying signal-propagation physics in a way that builds user intuition or supports algorithm development.

This project, carried out in collaboration with EDF, addresses this gap by developing an interactive simulation tool for cable fault detection. The mission objective is to provide an interface allowing simulations to be performed by running a pulse through a modeled cable and measuring the reflected signal. The tool models a cable as a distributed-parameter transmission line governed by the telegrapher's equations, introduces localized impedance discontinuities to represent faults, and simulates the resulting signal propagation and reflections in both the time and frequency domains. The scope is strictly limited to fault detection and localization: identifying where along the cable an anomaly occurs, rather than diagnosing its physical cause.

0.2 Research Question

The key success criteria are the accuracy of the fault localization, the ease of use and interactivity of the interface, and the numerical stability of the simulation. The tool is intended to be used not only by EDF engineers, but more broadly by any engineer or technician wishing to observe how signals behave in faulty cables and how detection methods compare, without requiring specialized signal processing expertise. This leads to the following research question:

To what extent can an interactive simulation tool, based on a transmission line theory, be used to reliably localize cable faults from reflected signal analysis while remaining accessible to users without signal processing expertise?

This report addresses this question by presenting the theoretical foundations, a review of popular detection methods, the numerical approach and implementation of the simulation tool, and the validation results obtained against known fault positions.

1. Theoretical Background

1.1 Transmission Line Theory

A transmission line is modeled as a distributed-parameter structure in which voltage and current vary as functions of both position and time, unlike ordinary circuit analysis, where a component's effect is treated as instantaneous everywhere along it, and only time matters. Whether position matters depends on how the cable's length compares to the signal's wavelength. At low frequencies, the wavelength is much

larger than the cable length, so the voltage can be treated as uniform along it at any instant : this is the quasi-static approximation, where ordinary circuit equations remain valid. This would also apply at higher frequencies if the cable is short enough relative to the wavelength. When the wavelength becomes comparable to the cable length, so different points along the cable can be at different phases of the signal at the same time. In this case, the distributed model is required, described by the telegrapher's equations developed below.

For the model to be valid, the propagation mode must be TEM (Transverse Electro-Magnetic) or quasi-TEM: both the electric field E and the magnetic field B are perpendicular to the direction of propagation. This assumption holds for coaxial cables and twisted pairs under typical operating conditions. Non-ideal effects such as imperfect conductors or inhomogeneous dielectrics introduce small corrections but can be put under the quasi-TEM approximation, if their influence remains small.

1.2 The RLCG Distributed Model

Under the TEM assumption, a transmission line of length L can be decomposed into infinitesimal segments of length Δz . Within each such segment, the resistance, inductance, capacitance, and conductance of that small piece of cable are represented by an equivalent circuit with four parameters per unit length :

- R (Ω/m): resistance per unit length, accounting for Joule losses in the conductors;
- L (H/m): inductance per unit length, representing magnetic energy stored along the segment;
- C (F/m): capacitance per unit length, representing electrostatic energy stored between the conductors;
- G (S/m): conductance per unit length, accounting for dielectric losses between the conductors.

This model is shown in Figure 1. If these parameters are independent of position z , the line is said to be uniform; otherwise it is non-uniform. A fault in the cable manifests as a localized deviation in one or more of these parameters, producing an impedance discontinuity.

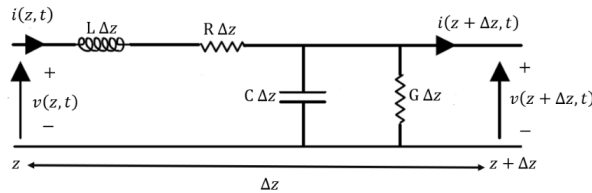


Figure 1. Equivalent circuit of Δz , from internship report

1.3 Derivation of the Telegrapher's Equations

Consider one of these small segments of length Δz , with parameters $R\Delta z$, $L\Delta z$, $C\Delta z$, and $G\Delta z$, as in Figure 1. To obtain the equations governing the line, we apply Kirchhoff's two laws to this segment and then let Δz shrink to zero.

1.3.1 Step 1 : Kirchhoff's Voltage Law (KVL)

Consider the loop formed by the top conductor of the segment (carrying R and L), the voltage $v(z, t)$ at the input, and the voltage $v(z + \Delta z, t)$ at the output. Going around this loop:

$$-v(z, t) + R(z)\Delta z(i(z, t)) + L(z)\Delta z \frac{\partial i(z, t)}{\partial t} + v(z + \Delta z, t) = 0$$

Rearranging and dividing both sides by Δz :

$$\frac{v(z, t) - v(z + \Delta z, t)}{\Delta z} = R \cdot i(z, t) + L \cdot \frac{\partial i(z, t)}{\partial t}$$

Taking the limit $\Delta z \rightarrow 0$, the left-hand side becomes:

$$\frac{\partial v(z, t)}{\partial z} = -R \cdot i(z, t) - L \cdot \frac{\partial i(z, t)}{\partial t}$$

1.3.2 Step 2 : Kirchhoff's Current Law (KCL)

At the node between the two conductors of the segment, some of the current $i(z, t)$ entering the segment leaks across to the other conductor through C and G , while the remainder, $i(z + \Delta z, t)$, continues onward. The current balance gives:

$$i(z, t) - G(z)\Delta z \cdot v(z + \Delta z, t) - C(z)\Delta z \cdot \frac{\partial v(z + \Delta z, t)}{\partial t} - i(z + \Delta z, t) = 0$$

Rearranging and dividing by Δz :

$$\frac{i(z + \Delta z, t) - i(z, t)}{\Delta z} = -G \cdot v(z + \Delta z, t) - C \cdot \frac{\partial v(z + \Delta z, t)}{\partial t}$$

Taking the limit $\Delta z \rightarrow 0$:

$$\frac{\partial i(z, t)}{\partial z} = -G \cdot v(z, t) - C \cdot \frac{\partial v(z, t)}{\partial t}$$

Together, these two coupled first-order partial differential equations:

$$\frac{\partial v(z, t)}{\partial z} = -R(z) \cdot i(z, t) - L(z) \cdot \frac{\partial i(z, t)}{\partial t}$$

$$\frac{\partial i(z, t)}{\partial z} = -G(z) \cdot v(z, t) - C(z) \cdot \frac{\partial v(z, t)}{\partial t}$$

are known as the telegrapher's equations. They fully describe how voltage and current evolve along the line as functions of both position z and time t .

Combining the two equations by differentiating and substituting one into the other decouples voltage and current, yielding the following second-order partial differential equations:

$$\frac{\partial^2 v(z, t)}{\partial z^2} = L \cdot C \cdot \frac{\partial^2 v(z, t)}{\partial t^2} + (R \cdot C + G \cdot L) \cdot \frac{\partial v(z, t)}{\partial t} + R \cdot G \cdot v(z, t)$$

$$\frac{\partial^2 i(z, t)}{\partial z^2} = L \cdot C \cdot \frac{\partial^2 i(z, t)}{\partial t^2} + (R \cdot C + G \cdot L) \cdot \frac{\partial i(z, t)}{\partial t} + R \cdot G \cdot i(z, t)$$

1.4 Sinusoidal Regime and Phasor Formulation

So far, the telegrapher's equations describe voltage and current as general functions of position and time. To make progress analytically, it is common to consider the special case of the sinusoidal regime; a signal oscillating at a single fixed frequency ω , after any initial transients have died out.

1.4.1 Phasor representation

A sinusoidal signal such as $v(z, t) = V(z) \cdot \cos(\omega t + \varphi(z))$ can be written compactly using a phasor: a complex number that encodes the amplitude and phase of the oscillation at each position z , with the time dependence $e^{j\omega t}$ factored out.

We write:

$$v(z, t) = \text{Re}(\bar{V}(z) \cdot e^{j\omega t}), \quad i(z, t) = \text{Re}(\bar{I}(z) \cdot e^{j\omega t})$$

where $\bar{V}(z)$ and $\bar{I}(z)$ are complex-valued functions of position only.

This turns the telegrapher's equations from partial differential equations (in z and t) into ordinary differential equations in z alone:

$$\frac{d\bar{V}(z)}{dz} = -(R + j\omega L) \cdot \bar{I}(z)$$

$$\frac{d\bar{I}(z)}{dz} = -(G + j\omega C) \cdot \bar{V}(z)$$

Differentiating one equation and substituting the other yields second-order ordinary differential equations for $\bar{V}(z)$ and $\bar{I}(z)$:

$$\frac{\partial^2 \bar{V}(z)}{\partial z^2} = \gamma^2 \bar{V}(z) \quad \frac{\partial^2 \bar{I}(z)}{\partial z^2} = \gamma^2 \bar{I}(z)$$

These equations have the general solutions:

$$\bar{V}(z) = \bar{V}^+ \cdot e^{-\gamma z} + \bar{V}^- \cdot e^{\gamma z}$$

and equation for I :

$$\bar{I}(z) = \bar{I}^+ e^{-\gamma z} + \bar{I}^- e^{\gamma z}$$

where γ is the complex propagation constant defined as:

$$\gamma = \sqrt{(R + j\omega L)(G + j\omega C)} = \alpha + j\beta$$

with $\alpha = \text{Re}(\gamma) = \text{Re}(\sqrt{(R + j\omega L)(G + j\omega C)})$, the attenuation constant and $\beta = \text{Im}(\gamma) = \text{Im}(\sqrt{(R + j\omega L)(G + j\omega C)})$ the phase constant.

The two terms represent two waves travelling in opposite directions: $\bar{V}^+ \cdot e^{-\gamma z}$ represents a wave travelling in the $+z$ direction (from source to load), and $\bar{V}^- \cdot e^{\gamma z}$ represents a wave travelling in the $-z$ direction (a reflected wave, travelling from the load back toward the source). The same form holds for the current $\bar{I}(z)$.

1.4.2 Phase velocity

The wavelength λ , defined as the distance over which the wave completes one full oscillation in space, is related to β by: $\lambda = \frac{2\pi}{\beta}$

The phase velocity v_ϕ is the speed at which a point of constant phase travels along the line, and is given by: $v_\phi = \frac{\omega}{\beta}$

1.4.3 Characteristic impedance

Finally, the ratio between the voltage and current phasors of a single travelling wave defines the characteristic impedance of the line: $Z_c = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$

This quantity represents the impedance "seen" by a wave as it propagates along an infinite line.

1.5 Special Cases

Depending on the parameters, special cases arise.

1.5.1 Lossless line

($R = G = 0$): the equations reduce to the ideal wave equation,

$$\frac{\partial^2 v(z, t)}{\partial z^2} = LC \cdot \frac{\partial^2 v(z, t)}{\partial t^2}$$

with phase velocity $v_\phi = 1/\sqrt{LC}$ and purely real characteristic impedance $Z_c = \sqrt{L/C}$.

1.5.2 Non-dispersive line

($R/L = G/C$): the signal propagates without distortion. The phase velocity remains $v_\phi = 1/\sqrt{LC}$ and $Z_c = \sqrt{L/C}$, but attenuation $\alpha = R\sqrt{C/L}$ is non-zero.

1.5.3 Low-loss line

($R \ll \omega L, G \ll \omega C$): the attenuation constant is approximated as

$$\alpha \approx \frac{\sqrt{LC}}{2} \cdot \left(\frac{R}{L} + \frac{G}{C} \right)$$

while the phase velocity and characteristic impedance remain close to their lossless values.

1.6 Reflection Coefficient and Fault Localization

When a propagating wave encounters an impedance discontinuity (such as a change in cable parameters caused by a fault, or a load at the line's termination) part of the wave is reflected back toward the source. The reflection coefficient at a load Z_L is defined as:

$$\Gamma_L = \frac{Z_L - Z_c}{Z_L + Z_c}$$

A matched load ($Z_L = Z_c$) gives $\Gamma_L = 0$ and no reflection. An open circuit ($Z_L \rightarrow \infty$) gives $\Gamma_L = +1$, and a short circuit ($Z_L = 0$) gives $\Gamma_L = -1$.

2. Numerical Methods : Time Domain (FDTD)

2.1 FDTD Discretization Framework

The telegrapher's equations are continuous in both space and time, but to simulate them numerically they must be discretized on a finite grid.

The Finite Difference Time Domain (FDTD) method replaces the continuous derivatives in the telegrapher's equations with finite differences over small, fixed intervals of space and time. The cable, of total length l , is divided into K segments of length $\Delta z = \frac{l}{K}$, and the simulation runs over a total duration t_{max} divided into N time steps of $\Delta t = \frac{t_{max}}{N}$, so that: $l = K \cdot \Delta z$, $t_{max} = N \cdot \Delta t$.

The cable parameters R, L, C, G are evaluated at each spatial grid point k : $R_k = R(k\Delta z)$, $L_k = L(k\Delta z)$, $C_k = C(k\Delta z)$, $G_k = G(k\Delta z)$.

This formulation naturally accommodates non-uniform lines, since each parameter can take a different value at each point k . This is what allows a localized fault to be represented.

The voltage and current are represented on a discrete grid U_k^n and I_k^n , where the superscript n denotes the time step and the subscript k denotes the spatial position.

The boundary and initial conditions of the simulation are:

- At the source ($z = 0$): $u(0, t) = u_s(t) - R_s \cdot i(0, t)$, where $u_s(t)$ is the injected signal and R_s is the source impedance.
- At the load ($z = l$): $u(l, t) = Z_L \cdot i(l, t)$.

Initial conditions:

- $u(z, 0) = 0$
- $i(z, 0) = 0$

for all z , i.e. the line is at rest before the signal is injected.

2.2 Stability and Accuracy Conditions

For the discretization to produce physically meaningful results, the spatial and temporal steps Δz and Δt cannot be chosen arbitrarily; they must satisfy the following conditions:

$$\Delta z \leq \lambda_{\min} \quad \Delta t \leq \frac{1}{f_{\max}} \quad \Delta t \leq \frac{\Delta z}{v_{\max}}$$

The first condition ensures that the spatial step is small compared to the shortest wavelength present in the signal, so that the voltage and current do not vary significantly within a single segment. The second and third conditions ensure that, within one time step, the signal cannot propagate further than one spatial step. If these conditions are violated, the simulation becomes numerically unstable, producing unphysical oscillations or divergence as we can see on Figure 2.

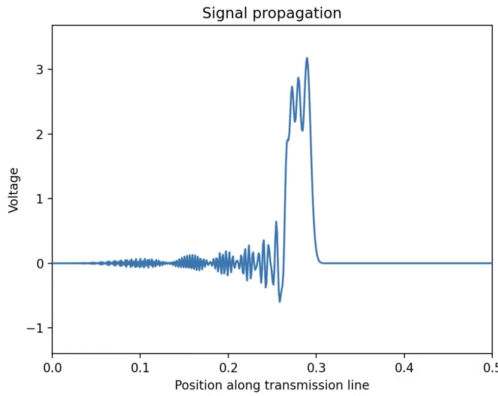


Figure 2. Example of Violation of Stability Conditions

2.3 First-Order Approximation

2.3.1 Interior points

Starting from the current equation:

$$\frac{\partial i(z, t)}{\partial z} = -G(z) \cdot v(z, t) - C(z) \frac{\partial v(z, t)}{\partial t}$$

The discretized form gives an update rule for the current at interior point k :

$$I_k^{n+1} = I_k^n \cdot \frac{L_k - R_k \Delta t}{L_k} - \frac{\Delta t}{L_k \Delta z} (U_{k+1}^n - U_k^n)$$

Similarly, starting from the voltage equation:

$$\frac{\partial v(z, t)}{\partial z} = -R(z) \cdot i(z, t) - L(z) \frac{\partial i(z, t)}{\partial t}$$

The discretized form gives an update rule for the voltage at interior point k :

$$U_k^{n+1} = U_k^n \cdot \frac{C_k - G_k \Delta t}{C_k} - \frac{\Delta t}{C_k \Delta z} (I_{k+1}^n - I_k^n)$$

These two equations allow the voltage and current at every interior point of the line to be updated from one time step to the next, using only values from the previous time step and from neighboring spatial points.

2.3.2 Source boundary condition

At the source ($k = 0$), the current is approximated by averaging its value over the time step:

$$I_S^n = \frac{U_S^{n+1} + U_S^n - U_0^{n+1} - U_0^n}{2Z_S}$$

Substituting this into the discretized current equation and solving for U_0^{n+1} gives the update rule at the source:

$$U_0^{n+1} = U_0^n \cdot \frac{(\Delta z \cdot 2Z_S)(C_k - G_k \Delta t) - \Delta t}{C_k \Delta z \cdot 2Z_S + \Delta t} + \frac{\Delta t}{C_k \Delta z \cdot 2Z_S + \Delta t} (U_S^{n+1} + U_S^n - 2Z_S I_1^n)$$

This equation couples the voltage at the source node to the injected source signal $U_S(t)$, the source impedance Z_S , and the current at the first interior point.

2.3.3 Load boundary condition

A similar procedure is applied at the load ($k = K$), where the current is given by:

$$I_L^n = \frac{U_K^{n+1} + U_K^n}{2Z_L}$$

Substituting into the discretized current equation gives the update rule at the load:

$$U_K^{n+1} = U_K^n \cdot \frac{(\Delta z \cdot 2Z_L)(C_k - G_k \Delta t) - \Delta t}{C_k \Delta z \cdot 2Z_L + \Delta t} + \frac{\Delta t \cdot 2Z_L I_K^n}{C_k \Delta z \cdot 2Z_L + \Delta t}$$

Together, these source and load conditions close the system, allowing the simulation to be advanced step by step over the full grid.

2.4 Second-Order Approximation

The first-order scheme, while simple to implement, introduces two distinct types of numerical error. First, because the scheme assumes the voltage and current remain constant within a single time and space step, this approximation itself introduces small artificial reflections. For instance, a parasitic reflection can appear at a matched load, even though physically no reflection should occur there, simply because the discretization treats the wave as piecewise-constant rather than continuous. Second, the discretization introduces numerical dispersion: the discrete scheme only approximates the true dispersion relation of the transmission line, so different frequency components of the signal travel at slightly different numerical phase velocities (an effect that depends on the chosen Δz and Δt). When a sharp step propagates,

this frequency-dependent speed distorts the pulse shape and produces the small oscillatory "ringing" trailing behind the front.

To reduce both effects, a second-order scheme is developed, in which the current is evaluated at half-integer time steps $(n + 1/2)$ and half-integer spatial points $(k + 1/2)$, and each update averages information from both directions in space and time. The discretized telegrapher's equations become:

$$\frac{U_{k+1}^n - U_k^n}{\Delta z} = -R_k \cdot \frac{I_{k-1/2}^{n+1/2} + I_{k-1/2}^{n-1/2}}{2} - L_k \cdot \frac{I_{k-1/2}^{n+1/2} - I_{k-1/2}^{n-1/2}}{\Delta t}$$

$$\frac{I_{k+1/2}^{n+1/2} - I_{k-1/2}^{n+1/2}}{\Delta z} = -G_k \cdot \frac{U_k^{n+1} + U_k^n}{2} - C_k \cdot \frac{U_k^{n+1} - U_k^n}{\Delta t}$$

As with the first-order scheme, these equations are applied separately at the source, at interior points, and at the load, to obtain update rules for each location on the grid.

2.4.1 Interior points

For interior points, the voltage update comes from the current equation:

$$U_k^{n+1} = U_k^n \cdot \frac{2C_k - G_k \Delta t}{2C_k + G_k \Delta t} - \frac{2\Delta t}{(2C_k + G_k \Delta t)\Delta z} \left(I_{k+1/2}^{n+1/2} - I_{k-1/2}^{n+1/2} \right)$$

and the current update comes from the voltage equation:

$$I_k^{n+1} = I_k^n \cdot \frac{L_k - R_k \Delta t}{L_k} - \frac{\Delta t}{L_k \Delta z} (U_{k+1}^n - U_k^n)$$

By symmetrizing the discretization in space and time, this scheme better approximates the continuous wave within each step, reducing parasitic reflections from the piecewise-constant approximation and improving reflection accuracy at genuine impedance discontinuities.

2.4.2 Source boundary condition

At the source, the current is approximated by averaging over the half-step:

$$I_S^{n+1/2} = \frac{U_S^{n+1} + U_S^n - U_0^{n+1} - U_0^n}{2Z_S}$$

Substituting into the discretized current equation and solving for U_0^{n+1} gives:

$$U_0^{n+1} = U_0^n \cdot \frac{(2C_k - G_k \Delta t)\Delta z Z_S - 2\Delta t}{(2C_k + G_k \Delta t)\Delta z Z_S + 2\Delta t} + \frac{2\Delta t (U_S^{n+1} + U_S^n - 2Z_S I_{1/2}^{n+1/2})}{(2C_k + G_k \Delta t)\Delta z Z_S + 2\Delta t}$$

2.4.3 Load boundary condition

At the load, the current is similarly given by:

$$I_L^{n+1/2} = \frac{U_K^{n+1} + U_K^n}{2Z_L}$$

Substituting into the discretized current equation and solving for U_K^{n+1} gives:

$$U_K^{n+1} = U_K^n \cdot \frac{(2C_k - G_k \Delta t)\Delta z Z_L - 2\Delta t}{(2C_k + G_k \Delta t)\Delta z Z_L + 2\Delta t} + \frac{4\Delta t Z_L I_{K-1/2}^{n+1/2}}{(2C_k + G_k \Delta t)\Delta z Z_L + 2\Delta t}$$

2.5 Comparison of the First- and Second-Order Methods

Both schemes were tested on identical simulation scenarios, with the same cable parameters and an impedance discontinuity along the line. The first-order scheme exhibits a small parasitic reflection at the load even when the load is matched to the line's characteristic impedance. This is due to the piecewise-constant approximation described before, and the small reflection can be seen in Figure 3, where a Gaussian signal was sent out to a matched load, and a small part of it was reflected back to the source.

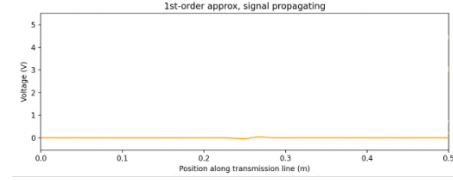


Figure 3. Small wave reflected from load impedance matched with characteristic impedance in our First Order Approximation.

The second-order scheme produces a noticeably cleaner result: when the load is matched, the parasitic reflection at the load is essentially eliminated, which can be seen in Figure 4. There, the same Gaussian pulse was sent out to the same matched load as in the first order, but this time (in the second order), none of the signal was reflected back to the source.

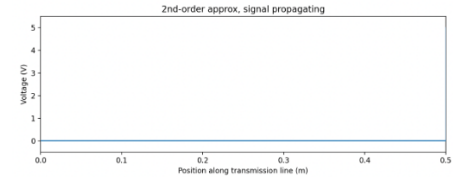


Figure 4. Nothing reflected from load impedance matched with characteristic impedance in our Second Order Approximation.

Overall, the second-order method is preferred when reflection accuracy is important, as is the case for fault localization, while the first-order method remains useful as a simpler base-line implementation.

2.6 Fault Modeling

A fault is modeled as a localized region of the cable over which one or more of the RLCG parameters deviate from their nominal values, producing a localized change in the characteristic impedance of the line. Since the discretization framework already allows R , L , C , and G to be defined independently at each spatial point k , a fault is introduced simply by modifying R_k , L_k , C_k , and/or G_k over the grid points corresponding to the faulted section of cable, while leaving the rest of the line uniform. At each grid point k , the local characteristic impedance is approximated from the local RLCG values: $Z_{c_k} \approx \sqrt{\frac{L_k}{C_k}}$

In the uniform sections of the line, Z_{c_k} is constant and equal to the nominal characteristic impedance of the cable. Within the faulted region, Z_{c_k} differs from this nominal value, creating an impedance discontinuity. This approach allows a wide range of fault types to be represented.

3. Frequency Domain Analysis

The phasor framework established in Section 1 yields the general forms $\bar{V}(z) = \bar{V}^+ e^{-\gamma z} + \bar{V}^- e^{+\gamma z}$ and $\bar{I}(z) = \bar{I}^+ e^{-\gamma z} + \bar{I}^- e^{+\gamma z}$, but leaves the integration constants $\bar{V}^+$, $\bar{V}^-$ and the position dependence of the reflection coefficient unspecified. The time-domain FDTD approach of Section 2 resolves the transient pulse and is well suited to two-way travel-time localisation, but yields limited direct information about the impedance state of the line as a function of position.

The frequency-domain approach developed here complements it by sweeping a single-frequency excitation across a bandwidth and extracting, at each frequency, the impedance profile and the standing-wave pattern. Section 3.1 derives the position-dependent reflection coefficient and the impedance transformation along the line. Section 3.2 incorporates reflections at both terminals to produce the complete steady-state voltage and current distribution. Section 3.3 develops the S_{11} -based fault-analysis framework used in the simulation tool.

3.1 Reflection Coefficient Profile and Impedance Transformation

As established in Section 1, the characteristic impedance Z_c gives the voltage-to-current ratio for a single travelling wave. For the combined forward and backward waves, the current phasor follows from the first telegrapher's equation as:

$$\bar{I}(z) = \frac{\bar{V}^+}{Z_c} e^{-\gamma z} - \frac{\bar{V}^-}{Z_c} e^{+\gamma z}$$

The minus sign in the reflected component arises because the backward wave carries power in the $-z$ direction, reversing the current sense relative to the voltage. This sign difference, absent from the symmetric notation $\bar{I}^+ e^{-\gamma z} + \bar{I}^- e^{+\gamma z}$ used in Section 1, is essential for computing the correct impedance ratio and the standing-wave patterns examined below.

3.1.1 Local reflection coefficient

Whereas Section 1 defined the reflection coefficient Γ_L at the load terminal, fault localization requires knowing the effective ratio of reflected to incident amplitude at every position z along the line. Defining this local reflection coefficient as $\Gamma(z) = \bar{V}^- e^{+\gamma z} / \bar{V}^+ e^{-\gamma z}$ and evaluating at the load ($z = \ell$) gives $\Gamma_L = (\bar{V}^- / \bar{V}^+) e^{2\gamma \ell}$. Substituting $\bar{V}^- / \bar{V}^+ = \Gamma_L e^{-2\gamma \ell}$ back yields:

$$\Gamma(z) = \Gamma_L \cdot e^{2\gamma(z-\ell)}$$

Since $z \leq \ell$ everywhere on the line, the exponent $2\gamma(z-\ell)$ has a non-positive real part, so $|\Gamma(z)| \leq |\Gamma_L|$ for all z : the effective reflection magnitude decays monotonically away from the

load, due to round-trip attenuation of the backward-travelling wave.

Factoring $\bar{V}^+ e^{-\gamma z}$ out of the voltage and current expressions then gives compact forms that cleanly separate the propagating contribution from the standing-wave modulation:

$$\bar{V}(z) = \bar{V}^+ e^{-\gamma z} (1 + \Gamma(z))$$

$$\bar{I}(z) = \frac{\bar{V}^+}{Z_c} e^{-\gamma z} (1 - \Gamma(z))$$

The factor $e^{-\gamma z}$ governs amplitude decay and phase shift from the source to position z . The modulation factor is $(1 + \Gamma(z))$ for voltage and $(1 - \Gamma(z))$ for current, reflecting the fact that the reflected wave adds constructively to the voltage but subtracts from the current.

3.1.2 Impedance transformation

The ratio $\bar{V}(z)/\bar{I}(z)$ gives the impedance looking into the line from position z toward the load:

$$Z(z) = \frac{\bar{V}(z)}{\bar{I}(z)} = Z_c \frac{1 + \Gamma(z)}{1 - \Gamma(z)}$$

Substituting $\Gamma(z) = \Gamma_L e^{-2\gamma x}$ (where $x = \ell - z$ is the distance from position z to the load) and simplifying via $\tanh(\gamma x) = (e^{\gamma x} - e^{-\gamma x}) / (e^{\gamma x} + e^{-\gamma x})$ yields the equivalent hyperbolic form, which is the practical formula for computing the impedance at any point given the load and line parameters:

$$Z(z) = Z_c \frac{Z_L + Z_c \tanh(\gamma x)}{Z_c + Z_L \tanh(\gamma x)}, \quad x = \ell - z \quad (1)$$

At $x = 0$ (the load terminal) this correctly returns Z_L . At $x = \ell$ ($z = 0$, the source terminal) it gives the *input impedance* seen by the source:

$$Z_{\text{in}} = Z(0) = Z_c \frac{1 + \Gamma_L e^{-2\gamma \ell}}{1 - \Gamma_L e^{-2\gamma \ell}} \quad (2)$$

Two canonical special cases follow directly. For a lossless *half-wavelength* line ($\alpha = 0, \beta \ell = n\pi$): $\tanh(j\beta \ell) = j \tan(n\pi) = 0$, giving $Z_{\text{in}} = Z_L$ regardless of Z_c , the line is electrically transparent to the source.

For a lossless *quarter-wavelength* line ($\beta \ell = \pi/2$): $\tanh(j\pi/2) \rightarrow \infty$, giving $Z_{\text{in}} = Z_c^2 / Z_L$ — the line inverts the load impedance.

3.1.3 Boundary conditions

At the source end, the voltage divider formed by the source impedance Z_s and Z_{in} sets the voltage injected onto the line:

$$V(0) = \frac{Z_{\text{in}}}{Z_s + Z_{\text{in}}} V_s$$

At the load end, Ohm's law must be satisfied: $V(\ell) = Z_L I(\ell)$. These two conditions, together with the reflection coefficient at the source terminal,

$$\Gamma_s = \frac{Z_s - Z_c}{Z_s + Z_c}$$

close the system at both ends. Like Γ_L defined in Section 1, Γ_s equals 0 for a matched source, +1 for an open circuit, and -1 for a short circuit. Together, Γ_L and Γ_s govern the multiple-reflection dynamics developed in Section 3.2.

3.2 Steady-State Distribution with Source and Load Reflections

Section 3.1 expressed $V(z)$ and $I(z)$ in terms of V^+ , which remained unresolved. When both terminals are mismatched ($\Gamma_s \neq 0$ and $\Gamma_L \neq 0$), a wave launched from the source reflects at the load, returns to the source, reflects again, and so on. Each round-trip introduces a multiplicative factor $\Gamma_s \cdot \Gamma_L \cdot e^{-2\gamma\ell}$. Summing the resulting geometric series in steady state:

$$\sum_{n=0}^{\infty} (\Gamma_s \Gamma_L e^{-2\gamma\ell})^n = \frac{1}{1 - \Gamma_s \Gamma_L e^{-2\gamma\ell}}, \quad |\Gamma_s \Gamma_L e^{-2\gamma\ell}| < 1$$

which converges whenever the round-trip attenuation exceeds the product of the reflection magnitudes (always satisfied for $\alpha > 0$). Applying the source boundary condition to resolve V^+ gives the complete steady-state voltage and current distributions:

$$V(z) = \frac{Z_c}{Z_c + Z_s} \cdot \frac{1 + \Gamma_L e^{2\gamma(z-\ell)}}{1 - \Gamma_s \Gamma_L e^{-2\gamma\ell}} \cdot V_s e^{-\gamma z} \quad (3)$$

$$I(z) = \frac{1}{Z_c + Z_s} \cdot \frac{1 - \Gamma_L e^{2\gamma(z-\ell)}}{1 - \Gamma_s \Gamma_L e^{-2\gamma\ell}} \cdot V_s e^{-\gamma z} \quad (4)$$

Each factor has a distinct physical role.

The term $V_s e^{-\gamma z}$ is the source voltage propagated to position z as if the line were infinite. The prefactor $Z_c/(Z_c + Z_s)$ (or $1/(Z_c + Z_s)$ for current) is the voltage divider at the injection point, setting the wave amplitude launched onto the line for a given source mismatch. The denominator $(1 - \Gamma_s \Gamma_L e^{-2\gamma\ell})$ accounts for the resonant build-up from all multiple reflections; as $\Gamma_s \Gamma_L e^{-2\gamma\ell} \rightarrow 1$ this term vanishes and the amplitude diverges, signalling a lossless-cavity resonance. Finally, the numerator factor $(1 \pm \Gamma_L e^{2\gamma(z-\ell)})$ combines the incident wave with the component reflected once at the load, producing the standing-wave modulation. The exponent $2\gamma(z - \ell)$ has a non-positive real part for $z \leq \ell$, correctly ensuring the reflected contribution decays as the observation point moves away from the load.

Equations (3)–(4) reduce to those of Section 3.1 when V^+ is identified as $V_s Z_c / [(Z_c + Z_s)(1 - \Gamma_s \Gamma_L e^{-2\gamma\ell})]$, confirming consistency between the two formulations.

These are the expressions evaluated numerically by the frequency-domain solver to compute voltage and current profiles at each frequency in the sweep.

3.3 Fault Localisation via the S_{11} Parameter

The distributions derived above give $V(z)$ and $I(z)$ at every interior point, but in a physical measurement only the input port is accessible. The diagnostically relevant quantity is the ratio of the reflected wave amplitude to the incident wave amplitude at the source terminal: the S_{11} parameter. In the scattering-matrix formalism, S_{11} is defined with respect to a reference impedance Z_{ref} (conventionally 50Ω) as:

$$S_{11}(f) = \frac{Z_{\text{in}}(f) - Z_{\text{ref}}}{Z_{\text{in}}(f) + Z_{\text{ref}}}$$

where $Z_{\text{in}}(f)$ is the input impedance from (2), computed from (R, L, C, G) through $\gamma(f)$ and $Z_c(f)$.

A localised fault at position d_f ; an impedance discontinuity $\Delta Z = Z_f - Z_c$, introduces an additional reflected wave that, after round-trip propagation over $2d_f$, contributes to S_{11} a term:

$$S_{11,\text{fault}}(f) \approx \Gamma_f \cdot e^{-2\gamma(f)d_f} = \Gamma_f \cdot e^{-2\alpha d_f} \cdot e^{-j4\pi f d_f / v_\phi}$$

where $\Gamma_f = (Z_f - Z_c)/(Z_f + Z_c)$ is the reflection coefficient at the fault. The amplitude factor $e^{-2\alpha d_f}$ captures the round-trip attenuation; the phase factor $e^{-j4\pi f d_f / v_\phi}$ causes $S_{11}(f)$ to oscillate sinusoidally in frequency at a rate proportional to the fault depth.

A fault at 30m therefore oscillates twice as rapidly as one at 15m, providing a direct spectral fingerprint of its position.

3.3.1 Differential analysis

Even without a fault, a physical cable produces a non-zero $|S_{11}(f)|$ due to connector reflections and residual source/load mismatch. To isolate the fault-specific contribution, the differential signature is computed:

$$\Delta S_{11}(f) = S_{11,\text{faulty}}(f) - S_{11,\text{healthy}}(f)$$

This subtraction cancels all contributions common to both configurations, preserving only the perturbation introduced by the fault. The module $|\Delta S_{11}(f)|$ is therefore a fault-only reflection metric: its amplitude is proportional to the severity of the impedance mismatch, and its oscillation rate in frequency encodes the fault depth through $\Delta f = v_\phi/(2d_f)$.

A resistive short ($\Gamma_f \rightarrow -1$) and an open circuit ($\Gamma_f \rightarrow +1$) produce equal-amplitude oscillations in $|\Delta S_{11}|$ but with opposite phase; a distinction recovered in the full complex ΔS_{11} and exploited in the impedance profile reconstruction below.

3.3.2 Spatial reflection profile

$\Delta S_{11}(f)$ acquired over a frequency sweep f_1 to f_N is the Fourier representation of the cable's fault impulse-reflection response. Applying the Inverse Discrete Fourier Transform (IDFT) recovers the time-domain fault response:

$$r(t) = \text{IDFT}\{\Delta S_{11}(f_k)\}$$

A reflection from a discontinuity at distance d_f arrives at the input port after a round-trip delay $t = 2d_f/v_\phi$. Converting $r(t)$ to a function of distance gives the *spatial reflection profile* $r(d)$, with a peak at $d = d_f$ localising the fault.

The spatial resolution of this profile is set by the measurement bandwidth $B = f_N - f_1$:

$$\Delta d_{\text{min}} = \frac{v_\phi}{2B}$$

A wider frequency sweep yields finer spatial resolution. In the simulation, the bandwidth is chosen so that Δd_{min} remains well below the 5% localisation accuracy target. A Hann window is applied to $\Delta S_{11}(f_k)$ before the IDFT to suppress side-lobe artefacts around the main reflection peak, reducing the risk of spurious peaks being misidentified as additional faults.

3.3.3 Impedance profile along the line

Beyond localization, the frequency-domain data permit reconstruction of the local impedance $Z(z)$ along the cable. The hyperbolic transformation of (1) is applied recursively, stepping backward from the load toward the source in spatial increments Δz :

$$Z_{\text{in}}(z, f) = Z_c \frac{Z_{\text{in}}(z + \Delta z, f) + Z_c \tanh(\gamma \Delta z)}{Z_c + Z_{\text{in}}(z + \Delta z, f) \tanh(\gamma \Delta z)}$$

Starting from Z_L at $z = \ell$ and iterating toward $z = 0$, this recursion yields the input impedance as seen from every position. The local reflection coefficient $\Gamma(z, f) = (Z_{\text{in}}(z, f) - Z_c) / (Z_{\text{in}}(z, f) + Z_c)$ then gives the effective local impedance:

$$Z_{\text{eff}}(z) = Z_c \frac{1 + |\Gamma(z, f_0)|}{1 - |\Gamma(z, f_0)|}$$

evaluated at a representative frequency f_0 .

A plot of $Z_{\text{eff}}(z)$ provides a spatially resolved fault signature: a localised dip indicates a resistive fault ($Z_f < Z_c$, partial short circuit), a localised peak indicates an inductive or open-circuit fault ($Z_f > Z_c$), and a gradual drift in the baseline suggests distributed degradation such as dielectric ageing or water ingress. The difference $Z_{\text{eff, faulty}}(z) - Z_{\text{eff, healthy}}(z)$ provides the same differential isolation as ΔS_{11} , highlighting the impedance perturbation introduced exclusively by the fault.

4. Implementation

Building on the theoretical framework developed in the previous sections; the telegrapher's equations, their FDTD discretization in first- and second-order forms, and the frequency-domain analysis of reflection and standing waves, this section describes how that theory was translated into a working simulation tool for cable fault localization. The goal of the tool is to let a user configure a cable (its RLCG parameters, length, and the presence of one or more localised faults), inject a test signal, and visualise how that signal propagates along the cable and how it appears when reflected back to the source, from which the position of a fault can be inferred.

The implementation is organized around the same building blocks introduced theoretically: a choice of test signal and its associated bandwidth requirements (Section 4.1), a description of the cable and any fault regions in terms of per-unit-length RLCG parameters (Sections 4.2-4.3), a numerical time-domain solver based on the FDTD update equations together with automated reflection-detection logic (Section 4.4), a frequency-domain visualization built on the impedance and reflection-coefficient formulation derived earlier (Section 4.5), a set of performance and stability considerations that keep the simulation both fast and numerically valid (Section 4.6), and finally the overall software architecture tying these components together into the Streamlit-based simulation tool (Section 4.7).

4.1 Signal Parameters

The injected test signal is a Gaussian pulse (in the time domain), chosen for its smooth, well-localized shape in both time and frequency. It is defined as:

$$V_s(t) = A \cdot \exp\left(-\frac{(t - t_0)^2}{2\sigma_t^2}\right)$$

where A is the pulse amplitude, σ_t is the pulse width in seconds, and t_0 is a time offset that centers the pulse away from $t = 0$ so that the full rising edge of the Gaussian is captured (in the implementation, $t_0 = 6\sigma_t$). The pulse width σ_t is the primary user-facing parameter, since it directly determines the bandwidth of the injected signal. A Gaussian pulse of width σ_t has an associated maximum frequency content:

$$f_{\text{max}} = \frac{1}{2\pi\sigma_t}$$

This relationship means that a narrower pulse (smaller σ_t) contains higher-frequency components, which in turn requires finer spatial and temporal discretization to resolve correctly. The corresponding minimum wavelength is:

$$\lambda_{\text{min}} = \frac{v}{f_{\text{max}}}$$

where v is the propagation speed on the (nominal) line, $v = 1/\sqrt{LC}$.

The points-per-wavelength parameter (PPW) acts as an accuracy knob: it sets how many spatial grid points are used to resolve the shortest wavelength present in the signal. A minimum of around 10 points per wavelength is recommended to avoid numerical dispersion; increasing PPW improves accuracy at the cost of a finer grid and therefore more computation.

4.2 Cable Parameter Controls

The cable is described by the four primary per-unit-length parameters of the RLCG transmission-line model: resistance R (/m), inductance L (H/m), conductance G (S/m), and capacitance C (F/m), together with the total physical length of the line. These four parameters determine both the characteristic impedance of the (nominal) line, $Z_0 = \sqrt{\frac{L}{C}}$ and the nominal propagation speed, $v = \frac{1}{\sqrt{LC}}$.

In the implementation, these reference values ($R_{\text{ref}}, L_{\text{ref}}, G_{\text{ref}}, C_{\text{ref}}$) are taken from the first cable section and used as the baseline against which any fault section is compared. The cable is represented internally as a set of per-node arrays ($R_{\text{arr}}, L_{\text{arr}}, G_{\text{arr}}, C_{\text{arr}}$), each of length N_x , so that every spatial grid point carries its own local RLCG values. This array-based representation is what allows a fault to be introduced as a localized change in these parameters without altering the overall structure of the solver.

4.3 Fault Configuration

A fault is modeled as one (or more) sections of the cable whose RLCG parameters differ from the nominal values over a finite stretch of the line. Each section is specified by a start

and end position (as fractions or absolute positions along the total length), together with its own R , L , G , C values:

$$\text{sections} = [(x_{\text{start}}, x_{\text{end}}, R_{\text{sec}}, L_{\text{sec}}, G_{\text{sec}}, C_{\text{sec}}), \dots]$$

For example, a purely resistive fault is introduced by raising R over a short interval while leaving L , G , and C at their nominal values (representing localized damage that increases series loss, such as corrosion or a degraded connector), whereas an impedance-mismatch fault (representing a change in cable geometry or dielectric) is introduced by changing L and/or C over a short interval, which locally shifts the characteristic impedance $Z_{\text{fault}} \approx \sqrt{\frac{L_{\text{sec}}}{C_{\text{sec}}}}$ away from Z_0 and produces a reflection.

The section boundaries are mapped onto the spatial grid using a boolean mask over the node-position array

$x = \text{linspace}(0, \text{length}, N_x)$: for each section, all grid nodes whose position falls within $[x_{\text{start}}, x_{\text{end}}[$ (with the last section's right edge inclusive) are assigned that section's R , L , G , C values. This produces the per-node arrays R_{arr} , L_{arr} , G_{arr} , C_{arr} used by the update equations.

The load termination at the far end of the cable is set equal to the nominal characteristic impedance ($Z_L = Z_0$), so that in the absence of a fault the line is matched and no reflection occurs from the far end beyond what is expected from the fault itself; any reflection seen at the source is therefore attributable to the fault section(s) introduced along the line.

4.4 Time-Domain Visualization

4.4.1 First-Order and Second-Order FDTD Solvers

Two FDTD solvers were implemented, corresponding to the first-order and second-order discretizations of the telegrapher's equations introduced in the theoretical section.

The first-order solver updates current and voltage on a co-located grid: at each time step, the interior currents $I[n+1, k]$ are advanced from the previous current $I[n, k]$ and the spatial difference of the voltage $V[n, k+1] - V[n, k]$, using a simple explicit forward-difference scheme.

The source and load boundary voltages are then updated from these new currents using the boundary-condition formulas derived for a generator of impedance Z_s and a load of impedance Z_l , and finally the interior voltages $V[n+1, k]$ are advanced from $V[n, k]$ and the spatial difference of the newly updated currents $I[n+1, k] - I[n+1, k-1]$.

Each of these four updates (interior current, source boundary, load boundary, interior voltage) is computed explicitly from quantities already known at time step n or from currents already updated at step $n+1$, giving a straightforward leapfrog-style update with first-order accuracy in time.

The second-order solver follows the same overall structure: interior currents updated first, then the source and load boundaries, then interior voltages, but uses the more refined second-order update coefficients.

In the implementation, this is reflected in how the per-node coefficient arrays are constructed: rather than the simple $(1 - R \cdot dt/L)$ and $(1 - G \cdot dt/C)$ factors of the first-order

scheme, the second-order scheme uses coefficients of the form

$$A_V = \frac{2C - G \cdot dt}{2C + G \cdot dt} \quad B_V = \frac{2 \cdot dt}{(2C + G \cdot dt) \cdot dx}$$

$$A_I = \frac{L - R \cdot dt}{L} \quad B_I = \frac{dt}{L \cdot dx}$$

for the interior voltage and current updates, and analogous symmetrized forms

$$\text{coef_S_first}, \quad \text{coef_S_second}$$

$$\text{coef_L_first}, \quad \text{coef_L_second}$$

for the source and load boundaries, derived from denominators of the form $(2C \pm G \cdot dt) \cdot dx \cdot Z + 2 \cdot dt$. These coefficients arise from averaging the spatial derivative over a half time-step (centering the update in time rather than using a one-sided difference), which is what gives the scheme its improved, second-order accuracy.

In both solvers, these per-node coefficient arrays are pre-computed once from the R_{arr} , L_{arr} , G_{arr} , C_{arr} arrays and the chosen dt , dx , rather than being recomputed at every time step, since they depend only on the (fixed) cable configuration and discretization.

4.4.2 Signal Propagation

For visualization, the full voltage array $V[n, :]$ is recorded at every time step n , giving a two-dimensional array of shape (N_t, N_x) : one row per time step, one column per spatial node. This is animated as a snapshot of voltage versus position along the cable, updated frame by frame, so the user can watch the Gaussian pulse travel along the line, partially reflect at the fault section(s), and partially transmit onward to the load.

4.4.3 Reflected Pulse Detection

In parallel with the propagation animation, the voltage at the source node, $V[:, 0]$, is plotted as a function of time. This is the signal an instrument connected at the injection point would actually measure: the outgoing pulse, followed (after a round-trip delay) by any reflections returning from discontinuities along the line.

To automatically identify these reflections, the implementation works by thresholding the magnitude of $V[:, 0]$ against a small fraction of its peak value (`threshold_factor`), then locating the rising and falling edges of the resulting boolean mask to segment the time trace into discrete "events": contiguous stretches where $|V[:, 0]|$ exceeds the threshold. The first such event corresponds to the emitted pulse itself; all subsequent events are candidate reflections. Several complementary methods are then used to convert each reflection event into a physical fault location:

- **Direct wave-speed method:** using the known nominal propagation speed v , the round-trip time to each reflection peak is converted directly to a distance via

$$d = \frac{v \cdot t_{\text{round trip}}}{2}$$

This is the most direct approach but assumes the fault region does not significantly alter the local propagation speed.

- **End-reflection calibration method:** the expected round-trip time to the far end of the cable is estimated as $2 \cdot \text{length}/v$, and the reflection event whose timing falls closest to this estimate is taken as the “end” reflection. All other reflection times are then expressed as a ratio of this end-reflection time, and distances are computed proportionally:

$$d = \text{length} \cdot \frac{t_{\text{reflection}}}{t_{\text{end}}}$$

This self-calibrates against the actual measured end-of-line echo rather than relying purely on the nominal speed, which helps absorb small discretization or modeling errors.

- **Fault-length (entry/exit pairing) method:** a propagating pulse always produces two reflections at a fault region: one at the entry and one at the exit, where the wave re-encounters the nominal impedance. These two reflections are most clearly distinguishable when the fault spans a significant length (rather than a single point discontinuity). The entry reflection coefficient is computed from the ratio of the reflected to emitted peak amplitudes, $\Gamma = \frac{V_{\text{entry}}}{V_{\text{emit}}}$ from which the local fault impedance is recovered via $Z_{\text{fault}} = Z_0 \cdot \frac{1+\Gamma}{1-\Gamma}$. Depending on whether the fault was introduced as a change in C or in L , the propagation speed inside the fault region (v_{fault}) is computed accordingly, and the predicted exit-reflection amplitude is computed as

$$A_{\text{exit}} = A_{\text{entry}} \cdot \left(-(1 - \Gamma^2) \right)$$

The implementation searches subsequent reflection events for one matching this predicted amplitude within a tolerance; once found, the time difference between entry and exit round-trip times, combined with v_{fault} , yields the physical length of the fault region. Later reflections consistent with internal reverberations inside the fault (echoes spaced at multiples of this same time difference) are flagged and excluded from being misidentified as separate faults.

The choice between these methods is made automatically depending on the cable configuration: if L and C are uniform across all sections (i.e., only R or G differ, as for a purely resistive fault), the time-ratio method is used; if L or C varies across sections (indicating an impedance-type fault with a measurable physical extent), the impedance-pairing method is used to additionally estimate the fault length.

4.5 Frequency-Domain Visualization

4.5.1 Overview and Motivation

The frequency-domain module complements the time-domain FDTD solver by approaching fault localization from a differ-

ent angle: rather than tracking a transient pulse as it propagates, it models the cable’s steady-state response across a sweep of sinusoidal frequencies and converts that response back into the distance domain to locate discontinuities.

This approach mirrors what a vector network analyzer would measure on a real cable, making it directly relevant to the EDF context where frequency-domain reflectometry (FDR) is a standard diagnostic technique.

The implementation is organized as a three-stage pipeline: computing the complex reflection coefficient $S_{11}(f)$ across the frequency band, transforming it into a distance-domain reflectogram via inverse Fourier transform, and detecting and reporting fault positions from the resulting peaks.

4.5.2 Computing S_{11} via ABCD Chain Matrices

The starting point is the complex source-port reflection coefficient $S_{11}(f)$, which encodes how much of an injected sinusoidal signal is reflected back to the source at each frequency. Rather than running a time-domain simulation, $S_{11}(f)$ is computed analytically for each tone in the frequency sweep using the ABCD chain-matrix formalism introduced in Section 3. The cable is partitioned into an ordered sequence of two-port elements: uniform line segments described by their local RLCG parameters, separated by lumped discontinuities where faults occur. For each uniform segment of length l , the ABCD matrix is evaluated analytically as:

$$M = \begin{bmatrix} \cosh(\gamma l) & Z_c \sinh(\gamma l) \\ \sinh(\gamma l)/Z_c & \cosh(\gamma l) \end{bmatrix}$$

where we recall that $\gamma = \sqrt{(R + j\omega L)(G + j\omega C)}$ is the propagation constant and $Z_c = \sqrt{(R + j\omega L)/(G + j\omega C)}$ is the local characteristic impedance, both evaluated at each frequency in the sweep.

Fault regions are incorporated by assigning locally modified RLCG values to the segments they occupy. An impedance-mismatch fault changes L or C over a finite stretch of the line, shifting the local Z_c away from its nominal value and producing a reflection there.

Hard faults (open or short circuits) are represented as limiting cases: an open circuit is modeled as a large lumped series impedance (giving $\Gamma \rightarrow +1$) and a short circuit as a large shunt admittance (giving $\Gamma \rightarrow -1$).

The total ABCD matrix of the full cable is obtained by cascading these element matrices in propagation order. From this total matrix, the input impedance at the source port is:

$$Z_{\text{in}}(f) = \frac{A \cdot Z_L + B}{C \cdot Z_L + D}$$

and the source-port reflection coefficient follows as:

$$S_{11}(f) = \frac{Z_{\text{in}} - Z_s}{Z_{\text{in}} + Z_s}$$

This calculation is performed twice: once for the healthy cable and once for the faulty cable, yielding a complex baseline difference $\Delta S_{11} = S_{11}^{\text{faulty}} - S_{11}^{\text{healthy}}$.

Subtracting in the complex domain rather than in magnitude

is essential: it is the phase of $S_{11}(f)$ that encodes the round-trip delay to each reflector and therefore its distance from the source, so subtracting only magnitudes would destroy this distance information entirely.

4.5.3 From Frequency Response to Distance Domain

The complex baseline difference $\Delta S_{11}(f)$ is converted to a distance-domain reflectogram by inverse Fourier transform. Before transforming, a window function is applied to $\Delta S_{11}(f)$ to control the trade-off between spatial resolution and sidelobe suppression. A rectangular window gives the finest theoretical resolution:

$$dx = \frac{v_p}{2B}$$

where $B = f_{\max} - f_{\min}$ is the total swept bandwidth and $v_p = 1/\sqrt{LC}$ is the lossless propagation speed. Smoother windows such as Hann or Blackman suppress sidelobes at the cost of broadening each peak by a window-dependent factor, so the effective resolution becomes:

$$dx_{\text{eff}} = w \times dx$$

where w is the window broadening factor. The windowed spectrum is zero-padded before transforming, which interpolates the distance-domain curve for a smoother visual appearance without improving the true resolution, which remains fixed by the bandwidth B regardless of the number of points used. The delay axis of the resulting impulse response is converted to a one-way distance axis using:

$$x = \frac{v_p \cdot t}{2}$$

where the factor of two accounts for the round trip: a reflection from a discontinuity at distance d returns after a travel time of $2d/v_p$ and therefore appears as a peak at $x = d$ in the distance domain. A second key metric follows from the frequency spacing df between tones: the maximum unambiguous range is

$$x_{\max} = \frac{v_p}{2 \cdot df}$$

beyond which reflections alias back to shorter apparent distances. The frequency sweep must therefore be designed so that $x_{\max}$ exceeds the cable length.

The displayed quantity is the normalized magnitude $|\Delta r(x)|$, whose peaks indicate the positions and relative strengths of reflections attributable to the fault. To localize the theoretical reflection strength at an isolated impedance step, the expected reflection coefficient is:

$$\Gamma = \frac{\Delta Z_c / 100}{2 + \Delta Z_c / 100}$$

where ΔZ_c is the percentage change in characteristic impedance at the discontinuity. This analytical value serves as a reference against which the height of detected peaks in $|\Delta r(x)|$ can be compared.

4.5.4 Ground-Truth Impedance Profile

Because the simulation has complete knowledge of the RLCG parameters at every point along the cable, it can compute the true local characteristic impedance:

$$Z_c(x, f_{\text{ref}}) = \sqrt{\frac{R(x) + j\omega L(x)}{G(x) + j\omega C(x)}}$$

evaluated at a user-specified reference frequency f_{ref} , and express it as a percentage deviation from the healthy-cable value:

$$\Delta Z_c(x) [\%] = 100 \cdot \frac{|Z_c(x, f_{\text{ref}})| - |Z_{c,\text{healthy}}|}{|Z_{c,\text{healthy}}|}$$

This ground-truth impedance profile is displayed alongside the reflectogram as a direct reference, making it possible to compare where the fault actually is and how strongly Z_c changes there against what the distance-domain peaks detect.

This profile is explicitly labeled as simulation ground truth; it is known exactly from the cable model and would not be accessible in a real measurement, where only $S_{11}(f)$ is observable.

4.5.5 Fault Localization and Resolution

Fault positions are estimated automatically by detecting peaks in $|\Delta r(x)|$ above a configurable prominence threshold and separated by a minimum distance.

The two key resolution limits : the theoretical rectangular-window resolution dx and the effective windowed resolution dx_{eff} , are computed from the sweep parameters and displayed directly on the reflectogram, giving a visual reference for what the measurement can and cannot resolve given the chosen frequency range and window.

A validation step then cross-checks detected peak positions against the true fault locations known from the simulation, reports the localization error for each fault, and flags whether pairs of closely spaced faults are separated by more than dx_{eff} and can therefore be resolved as distinct peaks, or whether they fall within the effective resolution and will merge into a single broad feature in the reflectogram.

4.6 Performance Considerations

4.6.1 Vectorized NumPy Operations

The update equations for voltage and current at interior nodes are applied to entire arrays at once using NumPy's vectorized array operations, rather than with explicit per-node Python loops. For example, the interior current update in the second-order solver is written as a single array expression:

$$\mathbf{I}[n+1, :] = \mathbf{A_I} * \mathbf{I}[n, :] - \mathbf{B_I} * (\mathbf{V}[n, 1:] - \mathbf{V}[n, :-1])$$

where $\mathbf{A_I}$ and $\mathbf{B_I}$ are the precomputed coefficient arrays derived once from the per-node R, L values and the time/space steps (rather than recomputed at every time step).

The same approach is used for the interior voltage update. Only the two boundary nodes (source and load) require special-case scalar update formulas, since they involve the external

source and load impedances. This vectorization is what makes it practical to run the simulation interactively: for a fixed number of time steps, the per-step cost scales as a small number of array operations over N_x points rather than N_x individual Python-level operations, which is significantly faster in an interpreted language.

4.6.2 Simulation Stability Enforcement

Two independent constraints govern the choice of time step dt , and the implementation enforces both automatically rather than leaving them to the user:

- **CFL (Courant-Friedrichs-Lewy) stability limit:** for an explicit FDTD scheme, the time step must satisfy

$$dt \leq \frac{dx}{v}$$

with a safety margin, taken here as $0.9 \cdot dx/v_{\max}$, using the fastest propagation speed among all sections to guarantee stability everywhere on the line. Violating this condition causes the numerical solution to grow without bound.

- **Nyquist sampling limit:** to faithfully represent the highest frequency content of the Gaussian pulse ($f_{\max}$), the time step must satisfy

$$dt \leq \frac{1}{2f_{\max}}$$

The implementation computes both limits (dt_{cfl} and dt_{nyq}) and takes $dt = \min(dt_{\text{cfl}}, dt_{\text{nyq}})$ guaranteeing that whichever constraint is tighter is respected. Diagnostic output reports both limits alongside the chosen dt and explicitly flags whether each constraint is satisfied, so that the consequences of a given choice of σ_i , PPW, and cable parameters are visible before running the full simulation. Similarly, the spatial step dx is derived from the shortest wavelength present (using the slowest propagation speed among sections, $v_{\min}$, to be conservative) divided by PPW:

$$dx = \frac{\lambda_{\min}}{\text{PPW}} = \frac{v_{\min}}{f_{\max} \cdot \text{PPW}}$$

and the number of spatial nodes N_x is rounded so that dx divides the total cable length exactly, ensuring the fault-section boundaries remain well-defined on the grid.

The total simulated time T_{total} is set based on the desired number of one-way or round trips (`n_trips`) at the length-weighted average propagation speed, so that reflections from both the fault and the line termination have time to return to the source before the simulation ends.

4.7 Software Architecture

4.7.1 Overview

The simulation tool is implemented as a Streamlit web application, chosen for its ability to combine a Python numerical backend directly with an interactive browser-based interface

without requiring a separate frontend framework.

The application is structured around two independent analysis modules: time-domain reflectometry and frequency-domain reflectometry. Both are accessible from a common homepage, reflecting the dual nature of the theoretical and numerical work described in the preceding sections.

The overall design goal was to make both methods accessible to a non-specialist user: all physically meaningful parameters are directly exposed and adjustable, results update immediately on demand, and the outputs are annotated sufficiently that the connection between the configured fault and the observed signal is self-evident.

4.7.2 Landing Page and Navigation

The homepage presents the two analysis modes side by side, each with a brief description of what it models and a button to enter that module, as shown in Figure 5. This separation reflects the fundamental difference in how the two methods work: the frequency-domain module computes the steady-state reflection coefficient analytically across a swept frequency band, while the time-domain module runs a full FDTD transient simulation. The user selects whichever mode is appropriate for the diagnostic question they want to explore and can return to the homepage at any time to switch between them.

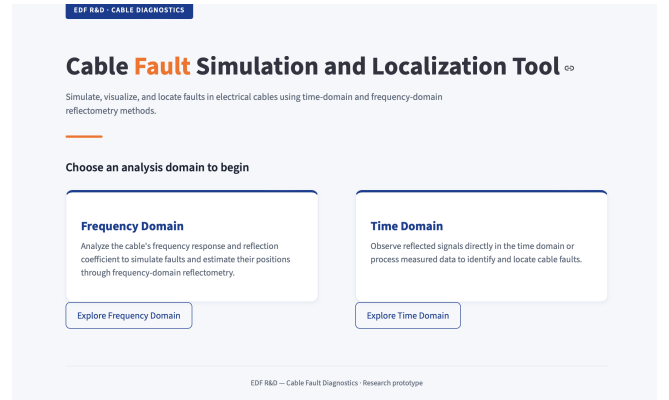


Figure 5. Homepage of the Simulation Tool

4.7.3 Time-Domain Module

The time-domain interface is organized into three sequential sections that mirror the logical structure of the simulation itself.

Section 1: Input signal

This lets the user choose between a Gaussian pulse and a sinusoidal excitation. For the Gaussian, the key parameter is $f_{\max}$ (in GHz), which sets the pulse bandwidth and therefore its spatial resolution on the cable. The amplitude (in volts) and points-per-wavelength (PPW) accuracy parameters are also adjustable here. A live input signal preview is displayed immediately as the user changes these parameters, showing the waveform in time before any simulation is run. For the Gaussian, it is a smooth bell-shaped pulse and for the sinusoidal option it is a continuous oscillation at the corresponding

frequency. This preview gives immediate feedback on the temporal character of the injected signal.

Section 2: Cable and fault

This exposes the physical cable configuration: total cable length in meters, the start and end positions of the fault region, the RLCG parameter being modified (R , L , G , or C , selected from a dropdown showing the healthy value in its label), and the fault multiplier that scales that parameter within the fault region. This maps directly onto the sections-based cable model described earlier: the healthy portion of the cable uses the default RLCG values, and the fault region uses the selected parameter multiplied by the given factor.

Section 3: Simulation

This contains the *Run simulation* button, which triggers the FDTD solver and produces two animated plots: the propagation snapshot showing voltage versus position along the cable at each time step (with the fault region highlighted in orange), and the TDR signal at the source node ($x = 0$) showing the full time trace of what would be measured at the injection point. A second static plot of the TDR signal at $x = 0$ is shown below the animation, giving a cleaner view of the reflected pulses without the animation updating. Below these plots, a simulated fault summary table reports the detected reflections: their peak voltage, round-trip time, estimated fault distance, and estimated spatial extent, computed using the reflection detection methods described before.

4.7.4 Frequency-Domain Module

The frequency-domain interface follows the same principle of sequential, logically grouped controls, organized into three labeled sections.

Section A: Cable configuration

This sets the physical cable length and the source and load impedances Z_s and Z_L . A collapsible panel labeled “Healthy cable model assumptions” allows the user to inspect or override the nominal RLCG values of the healthy cable.

Section B: Frequency sweep configuration

This exposes the three parameters that define the stepped-frequency excitation: start frequency, stop frequency (both in MHz), and the number of frequency samples. An information box reminds the user of the key physical relationships: the bandwidth $B = f_{\text{stop}} - f_{\text{start}}$ determines the spatial resolution $dx = \frac{v_p}{2B}$ while the frequency spacing $df = B/(N - 1)$ determines the maximum unambiguous range $x_{\text{max}} = \frac{v_p}{2 \cdot df}$. These two constraints mean that improving resolution requires a wider bandwidth, and extending the range requires more frequency samples over the same band.

Section C: Fault configuration

This allows the user to define one or more faults along the cable, each specified by its position, spatial length, modified RLCG parameter, and multiplication factor.

Multiple faults can be added or removed dynamically, enabling scenarios with several simultaneous discontinuities

whose separability relative to the sweep resolution can be directly explored.

Clicking *Run FDR Simulation* triggers the full pipeline described in Section 4.5: S_{11} is computed analytically for the healthy and faulty cables, the complex baseline difference is windowed and inverse-Fourier-transformed to produce the distance-domain reflectogram, peaks are detected, and results are displayed.

The output consists of a three-panel figure: the frequency-domain S_{11} response, the distance-domain reflectogram with detected peak markers and resolution bars, and the ground-truth impedance profile. These are followed by a simulation summary section showing key metrics (propagation velocity, bandwidth, number of detected reflections) and a results table listing each configured fault alongside its true region, the nearest detected peak position, and the localization error in meters. The ground-truth ΔZ_c column in this table makes explicit the difference between what the fault actually does to the cable impedance (known exactly from the simulation) and what the reflectogram is able to detect and localize.

5. Results and Validation

5.1 Results

The simulation tool successfully demonstrates fault detection across both analysis domains. In the time domain, the FDTD solver locates simulated faults with a position accuracy of approximately 1% of the total cable length, using the reflection detection methods described in Section 4.4.3.

This level of accuracy was achieved consistently across different fault types: resistive faults (elevated R), impedance-mismatch faults (modified L or C), and across different fault positions along the cable.

The frequency-domain module achieves comparable localization accuracy, with the additional advantage that fault positions can be read directly from a clean distance-domain reflectogram without requiring the time-consuming transient simulation that the FDTD approach demands.

A direct comparison between the first- and second-order FDTD solvers confirmed the theoretical expectation that the second-order scheme produces cleaner reflections with less numerical residue at the load, making the entry and exit reflections of an extended fault easier to distinguish.

For resistive faults, where L and C remain unchanged and the impedance mismatch is weak, the frequency-domain method proved markedly easier to use: the baseline subtraction $\Delta S_{11} = S_{11}^{\text{faulty}} - S_{11}^{\text{healthy}}$ isolates the fault contribution cleanly, whereas the time-domain approach requires careful threshold tuning to separate a small reflection from the noise floor of the emitted pulse. This comparison gives a practical sense of when each method is preferable and provides the groundwork for developing tools that could be applied to real cable measurements.

5.2 Validation

Validation of the simulation results was carried out along two complementary lines.

The first approach was literature-based. The theoretical foundations and reference implementations used throughout this work, including the FDTD discretization schemes, the ABCD chain-matrix formulation, and the reflectometry fault models, were drawn from several primary sources: an internship report from EDF, doctoral theses, and standard transmission-line textbooks.

In each case, the cable parameters and fault configurations described in these references were reproduced in the simulator, and the resulting reflectograms and localization outputs were compared against the published results. The agreement in reflection timing, amplitude, and spatial extent across these reference cases gave confidence that both the time-domain and frequency-domain implementations are physically correct.

In the case of our first and second order model simulations, which were heavily based on an internship report cited in our references, we decided to implement the parameters given in the report to observe how accurately our results matched with those of the intern. Figure 6 shows the expected result for the first situation of an inductance and capacitance fault in the first order simulation. Figure 7 shows our result for the same parameters, but a wire ten times shorter due to our simulation being long to run on longer wires due to the segmentation of the wire.

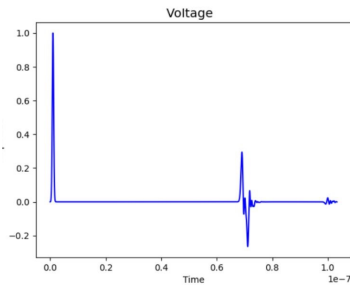


Figure 6. Voltage at point $x=0$ for First Order Approximation with a Capacitance and Inductance fault around the middle of the wire, taken from the internship report.

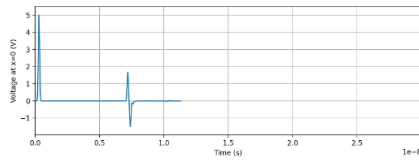


Figure 7. Voltage at point $x=0$ for First Order Approximation with a Capacitance and Inductance fault around the middle of the wire, from our simulation.

As we can see the results match both in shape and in timing, to a very close degree of precision, with the expected results.

We then moved on to validating both the first and second order model with those faults and a load impedance at the end of the wire which did not match the characteristic impedance

of the wire ($50\ \Omega$). Figure 8 shows the result obtained by the intern for a $10\ \Omega$ load resistance.

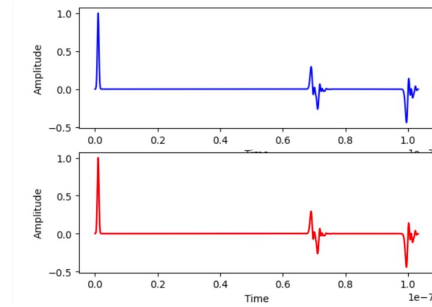


Figure 8. Voltage at point $x=0$ for First Order Approximation with a Capacitance and Inductance fault around the middle of the wire, and a load impedance of 10 Ohms, from the internship report.

The results that we obtained for this simulation can be seen in Figure 9 for the first order approximation and Figure 10 for the second order.

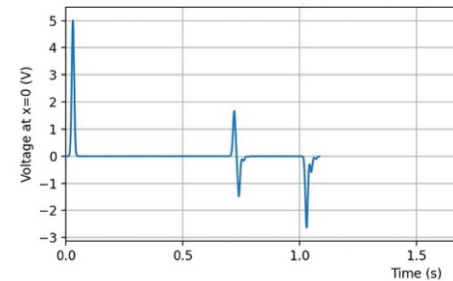


Figure 9. Voltage at point $x=0$ for First Order Approximation with a Capacitance and Inductance fault around the middle of the wire, and a load impedance of 10 Ohms, from our simulation.

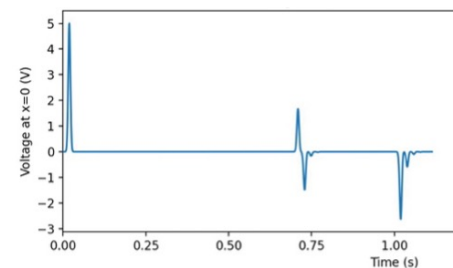


Figure 10. Voltage at point $x=0$ for Second Order Approximation with a Capacitance and Inductance fault around the middle of the wire, and a load impedance of 10 Ohms, from our simulation.

We then proceeded, like the internship report, to try this with a $100\ \Omega$ load resistance. Figure 11 shows the result obtained by the intern for a $100\ \Omega$ load resistance.

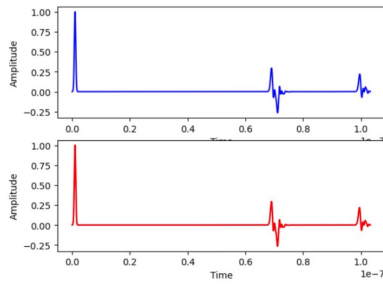


Figure 11. Voltage at point $x=0$ for First Order Approximation with a Capacitance and Inductance fault around the middle of the wire, and a load impedance of 100 Ohms, from the internship report.

The results that we obtained for this simulation can be seen in Figure 12 for the first order approximation and Figure 13 for the second order.

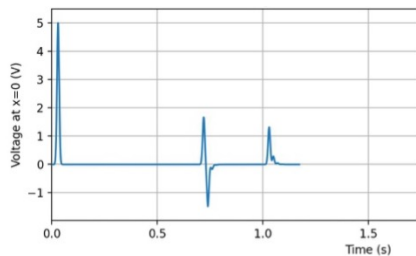


Figure 12. Voltage at point $x=0$ for First Order Approximation with a Capacitance and Inductance fault around the middle of the wire, and a load impedance of 100 Ohms, from our simulation.

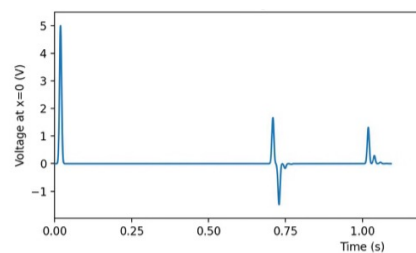


Figure 13. Voltage at point $x=0$ for First Order Approximation with a Capacitance and Inductance fault around the middle of the wire, and a load impedance of 100 Ohms, from our simulation.

These are only a few examples of the results that we cross-validated with the literature found on the subject, and all of them displayed consistency between our given output and the expected one, as can be seen in the Figures above. This allowed us to keep moving forward with our models to fine-tune them to the uses we had in mind for them, such as the interface and the localisation of faults.

The second validation approach was through direct collaboration with the industrial internship supervisor at EDF.

Throughout the development process, intermediate results, including signal propagation animations, reflectogram outputs, and fault localization tables were reviewed on an ongoing basis.

This feedback, grounded in practical experience with cable diagnostic systems, allowed errors and inconsistencies to be identified and corrected iteratively, and confirmed that the simulation behavior was consistent with what is observed on real cables in the field.

6. Discussion

The most crucial part of the project for us was to ensure the realisation of a tool that was both intuitive and genuinely useful. This simulation tool had to be easy to use and understand for people who didn't know much or even anything about the subject, but it also had to be accurate and give interesting results for experts in the field. This is why we decided to make an interface like the one we did. Unfortunately, our interface and our project as a whole has some limits.

The FDTD solver is a one-dimensional model based on the uniform transmission-line assumption. Real cables exhibit effects that this model does not capture: dielectric dispersion in the insulating material, connector reflections, and cross-talk between adjacent conductors in a bundle. Although certain methods such as FDR might still work accurately, finding a way of including these imperfect effects would help increase the accuracy and reliability of our diagnosis. The frequency-domain module assumes a perfectly known healthy baseline S_{11} healthy, which is used to compute the complex difference ΔS_{11} . In a real measurement, this baseline would itself be subject to noise, calibration errors, and slow drift in the cable's properties over time, all of which would degrade the localization accuracy. The current implementation has no noise model, so the accuracy figures reported here represent an upper bound on what could be achieved in a real measurement environment. The fault localization methods in the time domain depend on the propagation speed v_p being known. For the simulated cables this is exact ($v_p = \frac{1}{\sqrt{LC}}$), but in practice the velocity factor of a cable is known only approximately from the manufacturer datasheet and may vary with temperature and aging, introducing a systematic distance error proportional to the error in v_p . Finally, the tool currently models only a single straight cable section with at most a small number of isolated fault regions. Real electrical networks, particularly in industrial contexts such as EDF's infrastructure, are branched topologies with multiple junctions, varying cable types, and distributed aging. Extending the simulation to such networks, for instance using the scattering-matrix junction models described in the Ben Hassen thesis, would be a natural next step but is outside the scope of the present work.

7. Conclusion and Perspectives

7.1 Conclusion

This work set out to build a simulation tool for cable fault localization grounded in the physics of transmission-line theory, and to use it to study and compare the two principal reflectometry approaches, time-domain and frequency-domain in a controlled, pedagogically accessible environment. Both objectives were met. Starting from the telegrapher's equations and their FDTD discretization, two numerical solvers were implemented and validated: a first-order scheme and a more accurate second-order scheme, whose relative performance was directly observable through the tool's visualizations. A frequency-domain module was built in parallel, computing the complex reflection coefficient $S_{11}(f)$ analytically via ABCD chain matrices and converting it to a distance-domain reflectogram through inverse Fourier transform. Both modules were integrated into an interactive Streamlit interface that exposes all physically meaningful parameters to the user and presents results in a form accessible to both domain experts and newcomers to cable diagnostics. The tool successfully locates simulated faults with approximately 1% position accuracy in both domains.

The comparison between the two methods confirmed expected theoretical behavior: the frequency-domain approach isolates fault contributions cleanly through baseline subtraction and is particularly effective for weak impedance faults, while the time-domain approach provides a more direct and intuitive picture of signal propagation. It also gave concrete, quantitative grounding to trade-offs that are otherwise discussed only qualitatively in the literature. Both implementations were validated.

7.2 Perspectives

Several natural extensions of this work would significantly increase its practical value in the EDF context. The current tool models a single fault region per simulation in the time-domain module. Extending it to handle multiple simultaneous faults, with the associated challenges of overlapping reflections, entry/exit pairing ambiguity, and internal reverberations between closely spaced discontinuities would bring the simulator closer to the scenarios encountered on real cables, where degradation is rarely confined to a single isolated point. On the frequency-domain side, the integration of a full FDR mode including a realistic noise model and calibration errors would allow the tool to be used not just for ideal simulation but for sensitivity analysis: determining the minimum detectable fault severity as a function of bandwidth, frequency spacing, and signal-to-noise ratio, which is directly actionable information for designing real measurement protocols. A practically significant extension would be the ability to upload real oscilloscope or VNA measurements as input to the tool, replacing the simulated signal with a measured one and applying the same localization pipeline to actual cable data. This would allow the tool to be used as a diagnostic aid in the field, not just a simulation environment, and would close the loop between the theoretical work carried out here and the practical cable diagnostics work at EDF. Finally, and perhaps most

consequentially for EDF's longer-term objectives, the simulator could be used to generate large synthetic datasets of cable configurations and fault scenarios: varying cable length, RLCG parameters, fault type, position, severity, and number, paired with their ground-truth localization labels. Such datasets would provide a principled training basis for machine learning models aimed at automated fault diagnosis, contributing directly to EDF's AI development pipeline for cable health monitoring and avoiding the practical difficulties of acquiring and labeling sufficient real fault data.

Acknowledgments

We would like to thank the administration of the Bachelor of Global Engineering (BoGE) programme at CentraleSupélec for making this project possible. Organizing a technical project of this scope, across an international student cohort and in partnership with major industrial actors requires considerable effort and coordination, and we are appreciative of the care taken to ensure that students had the conditions necessary to carry out meaningful work.

We would also like to thank Samia Bouchafa-Bruneau, Academic Director of the Bachelor of Global Engineering, for her involvement throughout the project. Her mid-project feedback helped us take stock of where we stood and sharpen the direction of our work, and her broader role in structuring and overseeing the programme is what made this experience possible in the first place.

This project was carried out in partnership with EDF (Électricité de France), the French multinational energy company and one of the world's largest producers of electricity. We are grateful for the industrial context and support they provided, which gave this work both direction and real-world significance.

Finally, and above all, we would like to express our deepest gratitude to Susana Naranjo Villamil, our industrial supervisor at EDF, for her exceptional mentorship over the entire duration of this project. From the very beginning, she took the time to introduce us to the technical context, share her knowledge of cable diagnostics, and point us toward the right resources and references. She was consistently available, always willing to answer questions, and genuinely invested in our progress. Her feedback was clear, constructive, and delivered with the kind of enthusiasm that made every exchange both productive and encouraging. Her guidance shaped this project in fundamental ways, and working with her has been one of the highlights of this experience.

References

- [1] W. Ben Hassen, "Études de stratégies de diagnostic embarqué des réseaux filaires complexes," Doctoral thesis, Institut National Polytechnique de Toulouse, 2014.

- [2] EDF, “Simulation and analysis of transmission line parameters using FDTD and frequency-domain methods,” EDF, Internal internship report, Internal document provided by industrial supervisor at EDF.
- [3] Q. Zhang, “Efficient grid operation and maintenance with simulation apps,” *COMSOL Multiphysics*, pp. 24–26, Oct. 2018.

This hosted version can be used directly in any browser without cloning the repository or installing any dependencies. Note that it may be slower than the local version run from the repository, due to the limitations of the free hosting tier, but all features remain fully functional and the results are identical.

A. Code Repository and User Guide

All source code, simulation scripts, and documentation for this project are hosted in a Git repository accessible at the following address:

```
https://gitlab-student.centralesupelec.fr/lina.skik/edf_cablefault.git
```

Cloning the Repository

To obtain a local copy of the repository, open a terminal and run:

```
git clone https://gitlab-student.centralesupelec.fr/lina.skik/edf_cablefault.git
```

This will create a folder called `edf_cablefault` in your current directory containing all the project files.

Repository Structure

Once cloned, the repository contains the following:

- A `README.md` file at the root of the repository, which gives a full overview of the project structure, explains what each file and folder contains, and describes how the code is organised.
- The simulation source code for both the time-domain (FDTD) and frequency-domain modules.
- A user guide explaining how to navigate and use the Streamlit interface, including how to configure a cable, introduce a fault, run a simulation, and interpret the outputs.

Running the Interface

To launch the Streamlit interface locally, navigate into the cloned repository and follow the instructions in the `README`, which detail the required dependencies and the command to start the application. The interface runs in your browser and requires no additional installation beyond the Python dependencies listed.

Direct Online Access

For easier access, the interface is also available online at the following address, with no installation required:

```
https://cablefault-ehi3c5lqxdnre9q4bmhjii.streamlit.app/
```